

VisemeAR: Personalised Viseme Overlays for Korean Pronunciation Learning

Sia Lee

Sejong University
cyasiaseeya(at)gmail(dot)com

Abstract

Most pronunciation tools provide only numeric scores, leaving learners unsure how to adjust articulation. We present VisemeAR, a browser-based pronunciation trainer that overlays personalised vowel mouth shapes directly onto the learner's face. After a brief calibration with a neutral pose and three vowels (/a/, /u/, /i/), the system constructs a personal viseme space and generates individualised overlays for Korean monophthongs.

In a within-subject study with 24 beginner learners, items trained with personalised viseme overlays showed substantially larger pronunciation gains than score-only feedback (Cohen's $d = 0.94$). Observations of learner interaction revealed three recurring strategies—visual matching, contrast-based correction, and acoustic monitoring—indicating that the overlays influence not only learning outcomes but also learning behaviour.

Based on these findings, we propose a three-level design framework for AR pronunciation assistance: static visual guidance (Level 1), adaptive overlay modulation (Level 2), and vanishing overlays (Level 3) to encourage internalised articulatory control. Together, these results demonstrate that personalised articulatory visualisation can enhance webcam-based pronunciation training and inform the design of future AR learning interfaces.

Keywords

Augmented Reality, Pronunciation Learning, Visual Speech Feedback, Vowel Articulation, Personalised Calibration, Korean Language Learning

1. Introduction

Pronunciation feedback remains a central challenge in computer-assisted pronunciation training (CAPT), i.e., systems that use computational methods to support pronunciation practice and assessment in second-language learning. Modern automatic speech recognition systems such as Whisper [1] can provide pronunciation scores, yet learners receive only numeric feedback with little information on how to adjust their articulation. This gap between detecting an error and understanding how to correct it is especially salient for Korean vowels, where subtle

differences in mouth opening, lip rounding, and spreading produce distinct phonemic contrasts.

Visual articulatory cues can help bridge this gap: observing lip shapes and articulatory movements can facilitate second-language segmental learning [2, 3]. However, such systems are often impractical for everyday use, raising a key challenge for HCI: how can we provide accessible, personalised articulatory feedback without specialised equipment?

Recent work explores augmented reality (AR) as a deployable medium for articulatory guidance, suggesting that overlays can support engagement and perception of articulatory movement [8]. Yet existing AR tools use generic mouth diagrams, lack personalisation, and have rarely been evaluated under controlled conditions.

To address these limitations, we present VisemeAR, a browser-based pronunciation trainer that overlays a personalised mouth shape directly onto the learner's face during speech. After a brief calibration using /a/ (ㅏ), /u/ (ㅜ), and /i/ (ㅣ), the system constructs a personalised viseme space and generates overlays for Korean monophthongs, whose stable articulatory configurations support static visual targets. Unlike prior AR tools, VisemeAR adapts viseme landmarks to individual facial geometry via a calibrated 3D coordinate system.

Our contributions are threefold: (1) a personalised AR overlay system that adapts to facial geometry, (2) a lightweight calibration method using three vowels, and (3) a controlled within-subject evaluation showing that personalised viseme feedback improves Korean pronunciation beyond score-only training.

2. Related Work

2.1. Visual and Articulatory Feedback in CAPT

Visualising articulatory movements can support second-language pronunciation learning, with lip and jaw cues shown to enhance segmental perception [2] and multimodal speech processing [3]. Recent reviews of CAPT systems highlight the need for more accessible and personalised visual articulation feedback [5,6]. Yet little is known about how such personalised visualisations behave when delivered through deployable, webcam-only

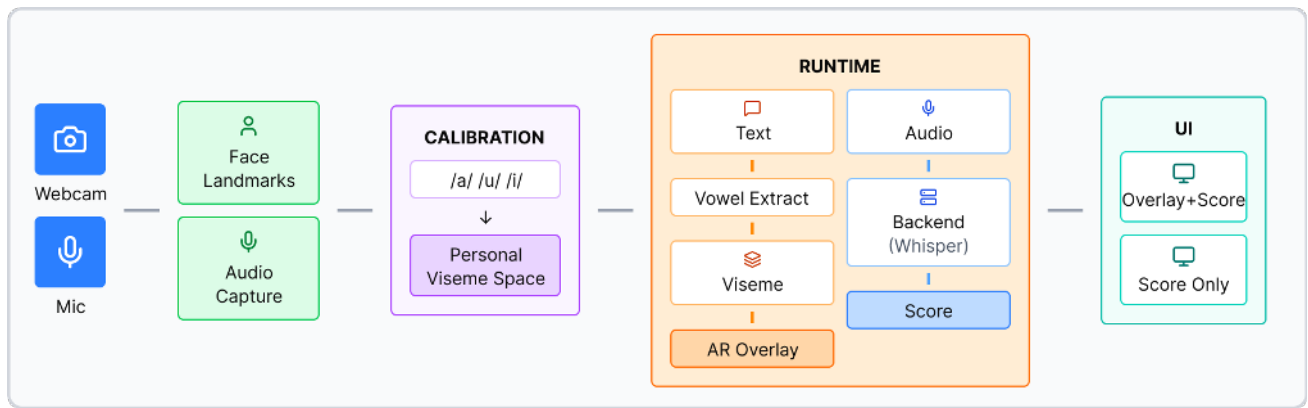


Figure 1 Overview of the VisemeAR system. A brief calibration of /a/, /u/, and /i/ is used to construct a personalised viseme space from lip landmarks. During runtime, the system extracts the target vowel from the displayed text, generates the corresponding viseme, overlays it on the learner’s face, and sends audio to a Whisper-based backend for scoring. The UI presents either Overlay+Score or Score-only feedback.

systems, especially when paired with ASR-based scoring in everyday CAPT contexts.

Most ASR-based CAPT tools offer only numeric scores; although Whisper provides high-quality transcription [1], such score-based feedback gives learners little insight into how to adjust articulation.

2.2. AR Interfaces for Pronunciation Training

Augmented reality has emerged as a promising medium for pronunciation training, increasing engagement and enabling situated feedback during speaking tasks. Systematic reviews similarly note benefits for contextualisation and motivation, alongside important design constraints [8]. However, most AR-based pronunciation tools still rely on generic mouth diagrams and rarely personalise feedback to the learner’s facial geometry; controlled evaluations are also scarce.

VisemeAR addresses these gaps by (1) generating user-specific viseme overlays from calibrated facial landmarks and aligning them with real-time head pose using only a webcam, and (2) conducting a controlled within-subject study that directly compares overlay-plus-score feedback to score-only training. To our knowledge, this is the first XR-based pronunciation trainer to personalise viseme shapes from user calibration and evaluate their impact in a controlled study of Korean vowels.

2.3. Korean Vowels, Visemes, and Visual Speech

Korean vowels exhibit clear articulatory dimensions—opening, rounding, and spreading—that make them well-suited to visual feedback. Research on Korean lip-reading defines viseme categories and proposes feature extraction methods for identifying vowel classes from lip motion [4], offering linguistic motivation for lip-based visualisation. Broader visual-speech literature similarly emphasises the importance of robust lip-

region representations for analysis and recognition tasks [7]. VisemeAR adopts this perspective but shifts from recognition to training: instead of deriving visemes from video, it constructs a personalised vowel space from calibrated samples of /a/, /u/, and /i/. This approach focuses on Korean monophthongs, which exhibit relatively stable articulatory configurations. Because Korean diphthongs involve rapid, less visually distinct transitions, we restrict our study to monophthongs, where static overlays provide clear and interpretable articulatory targets.

3. System Overview

Figure 1 provides an overview of the VisemeAR system pipeline, including vowel calibration, personalised viseme generation, real-time overlay rendering, and ASR-based pronunciation scoring. The system is a browser-based pronunciation trainer that overlays personalised vowel mouth shapes directly onto the learner’s face, providing real-time visual articulation guidance during speech. The system combines individualised viseme modelling with automatic pronunciation scoring to support both articulatory understanding and immediate performance feedback.

3.1. Design Goals

VisemeAR was developed around three goals:

G1: Personalised visual guidance. Provide a vowel mouth-shape overlay tailored to each learner’s facial geometry rather than generic diagrams.

G2: Low-friction deployment. Operate using only a standard webcam and microphone, without specialised sensors or head-mounted displays.

G3: Complementary scoring. Pair visual articulatory guidance with an objective pronunciation score to help learners track improvement.

3.2. Vowel Calibration and Personal Viseme Space

At first use, the system prompts learners to produce a neutral expression followed by three corner vowels—/a/ (ㅏ), /u/ (ㅜ), and /i/ (ㅣ). We extract 40 lip landmarks (20 outer, 20 inner contour) using MediaPipe Face Mesh, then apply a two-stage stabilisation procedure (temporal exponential smoothing and head-pose normalisation). Figure 2 illustrates the outer (green) and inner (magenta) lip landmarks extracted during calibration to construct the personalised viseme space.

A personal coordinate system is defined from the neutral frame: the origin is the midpoint of the upper- and lower-lip centres, with axes derived from the inter-ocular direction and nose-forehead vector. This frame provides a stable reference for modelling lip shapes.

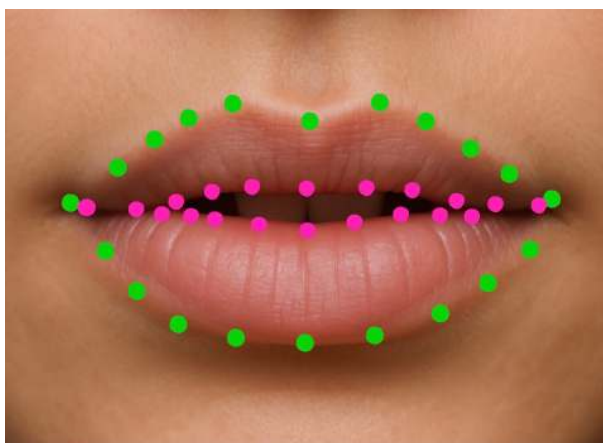


Figure 2 Outer (green) and inner (magenta) lip landmarks used by VisemeAR to build a personalised viseme space during the calibration step.

The three calibrated vowels serve as articulatory basis vectors representing mouth opening (/a/), rounding (/u/), and spreading (/i/). Each monophthong is associated with interpolation coefficients over these dimensions, allowing us to generate individualised target shapes. This linear combination corresponds to the articulatory structure of Korean monophthongs, which can be approximated along the dimensions of opening, rounding, and spreading.

3.3. Vowel Overlay Generation

For each lip landmark, VisemeAR computes its displacement from neutral for /a/, /u/, and /i/, then applies a weighted combination based on the vowel's coefficients. Adding this displacement to the neutral position yields the target 3D landmark, producing a personalised viseme that reflects the learner's lip proportions.

At runtime, we estimate a dynamic coordinate system from live facial landmarks and transform the target shape into this frame. The overlay is scaled according to face size and aligned with head pose, ensuring stable rendering during practice.

3.4. Pronunciation Scoring and Feedback

Pronunciation accuracy is evaluated using a Whisper-based scoring pipeline [1] built on an open-source AI pronunciation trainer [9]. When the learner reads an item aloud, the browser records audio and sends it with the expected text to a serverless backend. The backend decodes and resamples the audio to 16 kHz, processes it through the trainer, and returns a global pronunciation score (0–100) and alignment information. The scoring model is treated as a black box; our study isolates the effect of adding personalised viseme guidance.

In Overlay+Score, both the overlay and score are shown. In Score-only, only the score is displayed.

3.5. Interaction Flow

During each training trial, VisemeAR displays a Korean syllable or word and, when enabled, the corresponding vowel overlay. The learner reads the item aloud, after which the system computes and displays a pronunciation score. Figure 3 shows the VisemeAR interface during calibration and practice, including real-time viseme overlays and pronunciation score feedback. Learners may retry items, linking real-time visual guidance with immediate performance feedback.

4. Experiment

4.1. Design and Hypotheses

We conducted a within-subject experiment with two feedback conditions: Overlay+Score and Score-only. In both conditions, learners received an automatic pronunciation score based on Whisper [1]; in Overlay+Score they additionally saw the personalised vowel mouth-shape overlay while speaking. All participants completed a pre-test and post-test without feedback.

We tested two hypotheses:

H1 (Effectiveness): Items trained with Overlay+Score will show larger pronunciation gains from pre- to post-test than items trained with Score-only.

H2 (Perceived usefulness): Learners will rate the overlay condition as more helpful and easier to understand than score-only feedback.

4.2. Participants

We recruited 24 adults (13 female, 11 male, age 19–25) who were true beginners in Korean pronunciation: all could read Hangul but had no formal pronunciation training, no prior speaking

courses, and no experience with AR or pronunciation-learning tools. Participants reported diverse ethnic and national backgrounds, including Indian (n = 13), Indonesian (n = 5), French (n = 3), American (n = 2), and Saudi (n = 1). All self-rated their Korean speaking proficiency as “Beginner,” and none reported speech or hearing disorders. Sessions were conducted individually in a controlled environment using the same laptop webcam (1080p, 30 fps), USB microphone, and diffuse LED lighting; background noise was kept below 40 dB. Five participants wore glasses, but landmark jitter remained minimal (<2 px). Seating distance (50–60 cm) and instructions were standardised across participants, who received a coffee voucher for compensation.

4.3. Stimuli

We used 24 CV syllables and 16 common words as stimuli. Table 1 summarises the syllable and word items used in the experiment, all of which contain Korean monophthong vowels.

Table 1. Syllable and word stimuli used in the experiment. All items contain Korean monophthong vowels only.

Type	Items (Korean)
Syllables (24)	가, 나, 다, 구, 누, 두, 기, 니, 디, 게, 내, 대, 게, 네, 데, 거, 너, 더, 고, 노, 도, 그, 느, 드
Words (16)	바다, 나라, 우산, 구두, 비누, 기린, 개미, 내일, 세모, 메모, 번호, 거미, 고기, 노트, 그림, 흐름

All items contained monophthong vowels only. Diphthongs such as /wa/ (ㅘ) or /wi/ (ㅚ) were excluded because their contrasts depend strongly on temporal trajectories and coarticulation, which are not well represented by a single static lip target. Syllables were split into two matched lists (S1, S2) of 12 items each, and words into two lists (W1, W2) of 8 items each, balanced for vowel distribution. For each participant, one syllable list and one word list were assigned to Overlay+Score, and the complementary lists to Score-only; this mapping was counterbalanced across participants.

4.4. Procedure

Each session lasted about 45–60 minutes.

Calibration and practice. After consent and a short background questionnaire, participants completed a one-time calibration in which they produced a neutral expression followed by /a/ (ㅏ), /u/ (ㅜ), and /i/ (ㅣ) to construct their personal viseme space (Section 3). A brief practice block with additional

items familiarised them with reading aloud, viewing the overlay, and receiving scores.

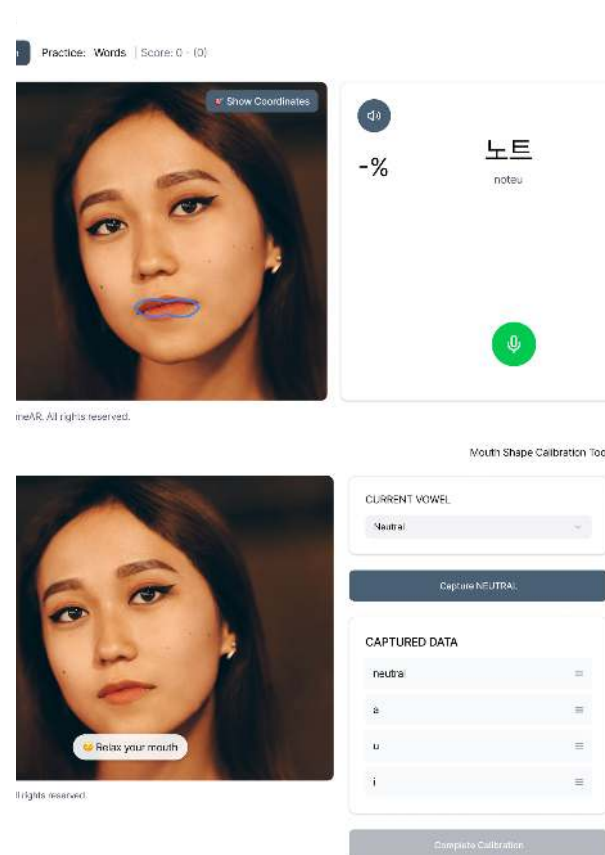


Figure 3 Interface views from VisemeAR. The top screenshot shows the calibration screen where learners capture neutral and vowel samples. The bottom screenshot shows the practice screen with real-time overlay and pronunciation score. The face shown is AI-generated and not associated with any participant.

Pre- and post-test (no feedback). In both pre- and post-test, participants read all syllables and words once each in random order. The interface displayed only the text; no overlay or scores were shown. Audio was recorded for later scoring.

Training phase. During training, only items assigned to Overlay+Score and Score-only were presented, in pseudo-random order with both conditions interleaved. For each item, the text appeared on screen; in the Overlay+Score condition, the corresponding vowel overlay was shown, while in Score-only it was hidden. Participants read the item aloud, the system sent the audio to the backend, and a pronunciation score (0–100) was displayed. Participants could attempt each item up to three times, with the best score shown. At the end of the session, participants completed a short Likert-scale questionnaire comparing the two feedback types and provided optional free-form comments.

4.5. Measures and Analysis

Objective measures. For each participant and item, we obtained pronunciation accuracy scores at pre-test and post-test from the Whisper-based scoring system. We defined the gain as the difference between post-test and pre-test scores. Items were labeled by their training condition (Overlay+Score vs. Score-only). For each participant, we averaged gains over items in each condition and compared the means using paired t-tests.

Subjective measures. Questionnaire responses (7-point Likert scales) on helpfulness, ease of understanding, and distraction were summarised per condition and compared using non-parametric tests. We additionally used open-ended comments to qualitatively illustrate how learners experienced the overlay.

5. Results

5.1. Pronunciation Accuracy

For each participant and item, we computed the gain as the difference between post-test and pre-test scores, then averaged gains separately for items trained in the Overlay+Score and Score-only conditions.

On average, items trained with Overlay+Score showed larger gains ($M = 9.8$, $SD = 4.3$) than items trained with Score-only ($M = 4.1$, $SD = 3.5$). A paired-samples t-test confirmed that this difference was statistically significant, $t(23) = 4.62$, $p < .001$, $d = 0.94$.

This pattern supports H1, indicating that the personalised viseme overlay led to larger improvements in pronunciation accuracy than score-only feedback as measured by this Whisper-based scoring model; future work will validate these gains against expert ratings and alternative ASR systems.

We also checked for baseline differences between item sets. Mean pre-test accuracy did not differ reliably between items later assigned to Overlay+Score and Score-only (Overlay+Score: $M = 72.4$, $SD = 6.9$; Score-only: $M = 71.7$, $SD = 7.2$; $t(23) = 0.41$, $p = .68$), suggesting that counterbalancing successfully controlled for initial difficulty. Table 2 summarises mean pre-test and post-test pronunciation scores across conditions, illustrating the larger gains observed for items trained with personalised viseme overlays.

Table 2. Mean pronunciation scores (SD) at pre-test and post-test for items trained with Overlay+Score and Score-only feedback.

Condition	Pre-test Mean(SD)	Post-test Mean (SD)	Gain
Overlay + Score	72.4 (6.9)	82.2 (6.1)	9.8
Score-only	71.7 (7.2)	75.8 (6.8)	4.1

5.2. Syllables vs. Words

We next examined syllables and words separately. For syllable items, mean gains were $M = 10.6$, $SD = 4.7$ in the Overlay+Score condition and $M = 5.1$, $SD = 3.9$ in the Score-only condition, $t(23) = 4.08$, $p < .001$.

For word items, mean gains were $M = 8.9$, $SD = 4.2$ (Overlay+Score) versus $M = 3.2$, $SD = 3.4$ (Score-only), $t(23) = 4.03$, $p < .001$.

Although the effect size was slightly larger for words (Cohen's $d = 0.82$ for syllables, $d = 0.83$ for words), both item types showed a consistent advantage for the overlay condition. These results suggest that the viseme-based feedback generalises from isolated syllables to simple words composed of the same monophthong vowels.

5.3. Subjective Feedback

Participants' ratings on 7-point Likert scales further supported the usefulness of the overlay. Learners rated the Overlay+Score condition as more helpful for understanding how to shape their mouth ($M = 5.8$, $SD = 0.9$) than the Score-only condition ($M = 4.1$, $SD = 1.2$). A Wilcoxon signed-rank test indicated that this difference was significant ($Z = 3.21$, $p = .001$), supporting H2.

Participants also reported that the overlay made it easier to notice differences between their own pronunciation and the target (Overlay+Score: $M = 5.6$, $SD = 1.0$; Score-only: $M = 4.3$, $SD = 1.3$; $Z = 2.89$, $p = .004$). Ratings of distraction were relatively low in both conditions (Overlay+Score: $M = 2.7$, $SD = 1.2$; Score-only: $M = 2.4$, $SD = 1.1$), and the difference was not statistically significant ($Z = 1.02$, $p = .31$), suggesting that the overlay did not substantially increase perceived cognitive load.

Open-ended comments were generally positive. Several participants mentioned that seeing the target lip shape "made the vowel feel clearer" or "helped me fix my mouth position," whereas a few noted that they occasionally focused more on matching the overlay than on listening to their own voice. Overall, the qualitative feedback aligned with the observed gains in pronunciation accuracy.

6. Discussion

Our findings provide initial evidence that viseme-based overlays can support Korean pronunciation learning. Items trained with Overlay+Score showed substantially larger gains than score-only items, suggesting that adding visual articulatory cues on top of numeric scores can help learners adjust their production. Subjective ratings further support this interpretation: learners found the overlay clearer and more helpful, and did not report increased distraction.

6.1. Interaction Observations & Design Framework

Participants employed three noticeable strategies during practice: visual matching, contrast-based correction, and acoustic monitoring. While static overlays helped learners understand how to shape their mouth, two participants briefly over-relied on matching the overlay, at times producing an incorrect vowel when shapes appeared visually similar, suggesting that visual mimicry can override phonemic intent. To reduce imitation and encourage internalisation, we propose a three-level design framework for AR pronunciation assistance. Level 1 (static overlay), implemented in VisemeAR, exposes a clear articulatory target and supports early visemic awareness but may encourage visual matching. Level 2 (adaptive overlay) modulates the salience or precision of the overlay (for example by shrinking, simplifying, or fading it as performance improves) to shift attention toward internal control. Level 3 (vanishing overlay) removes visual guidance and retains only acoustic or score-based feedback to probe retention and transfer. This progression aligns with scaffolding approaches in motor learning and suggests that AR pronunciation tools should transition from intensive visual guidance (Level 1) toward more autonomous production (Levels 2 and 3) over time.

6.2. Limitations & Future Work

This study focused on Korean monophthongs, which have stable articulatory targets well suited to static overlays; diphthongs and consonants may require time-varying viseme trajectories or multimodal feedback. Pronunciation accuracy was evaluated with the same Whisper-based system used for training. In this study, we intentionally treat Whisper as a black-box scorer and keep the acoustic model fixed in order to isolate the interface question of how different feedback modalities (numeric scores vs viseme overlays) shape learning behaviour. This design choice means that improvements could still be partially biased toward model-specific preferences, so future work should incorporate independent measures such as expert ratings, other ASR systems, or targeted acoustic analyses. Our evaluation used short, controlled

items in read speech, with beginner adult learners; further studies are needed to assess retention, transfer to new words, and applicability to more diverse learner groups. Finally, although we observed interaction strategies qualitatively, we did not systematically analyse articulatory changes (e.g., landmark shifts pre-post). Future work could quantify movement toward target configurations and examine how learners integrate visual and acoustic cues over time.

Despite these limitations, VisemeAR demonstrates that lightweight, browser-based articulatory feedback can enhance pronunciation learning without specialised hardware and offers a foundation for extending AR-based pronunciation training to dynamic visemes and mobile contexts.

7. Conclusion

This paper introduced VisemeAR, a personalised AR-based pronunciation training system that overlays user-specific vowel mouth shapes during speech production. By combining a calibrated viseme space with Whisper-based scoring, the system provides clear articulatory targets while remaining lightweight and fully browser-based. In a within-subject study with beginner Korean learners, the overlay condition produced significantly larger pronunciation gains than score-only feedback and was perceived as more helpful and intuitive. These findings highlight the promise of personalised articulatory visualisation and a design framework in computer-assisted pronunciation training and motivate future work on dynamic viseme trajectories, broader phonetic coverage, and real-world mobile deployment. To support reproducibility, we provide the implementation repository at <https://github.com/cyasiaseeya/VisemeAR>.

8. Acknowledgements

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